Linear programming

Linear programming

- ***find the conditions that optimize a function given some constraints.***
- ***the function and constraints can be expressed as linear equations or inequalities.***

Farmer’s problem

How to maximise the profit?

- 110 hectares of land
- Budget of $ 10.000
- Man-days: 1200
- Entire production can be sold

Variety	Cost (Price$/Hec)	Profit (Price$/Hec)	Man-days/Hec
Wheat	100	50	10
Barley	200	120	30

Terminology

- **Objective Function:** The function to be optimised
  - the profit
- **Decision (also termed Design) Variables:** influence the objective
  - hectares with wheat and barley
- **Constraints:** restrict or limit the decision variables.
  - available resources, i.e. area, budget and man-days.
- **Non-negativity restriction:** The decision variables should always be greater than or equal to 0.
  - areas with wheat and barley can only be positive.

Linear problem of the farmer

- **Step 1: Decision variables**
- The total area for growing Wheat = $X_1 \geq 0$ (in hectares)
- The total area for growing Barley = $X_2 \geq 0$ (in hectares)

- **Step 2: The objective function**
- net profit of $50 for each hectare of Wheat and $120 for each Barley.
- Our objective function **to be maximized is:** $f(X_1, X_2) = 50X_1 + 120X_2$

Step 3: The constraints

1. Budget of \$10,000

$$100\ X_1 + 200\ X_2 \leq 10000 \Rightarrow X_1 + 2\ X_2 \leq 100$$

2. 1200 man-days distributed between the crops

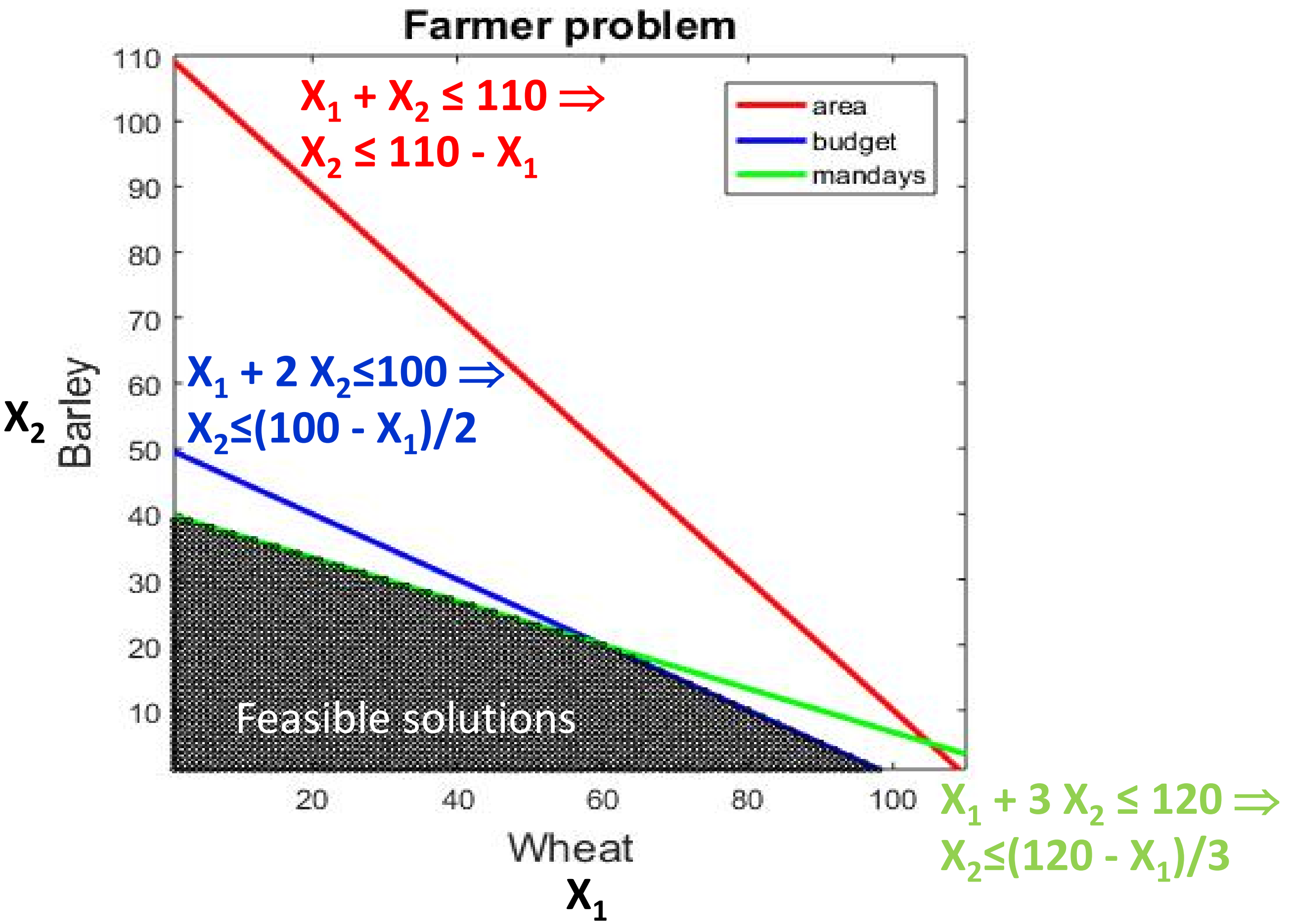
$$10\ X_1 + 30\ X_2 \leq 1200 \Rightarrow X_1 + 3\ X_2 \leq 120$$

3. Total area is 110 hectares

$$X_1 + X_2 \leq 110$$

Variety	Cost (Price\$/Hec)	Profit (Price\$/Hec)	Man-days/Hec
Wheat	100	50	10
Barley	200	120	30

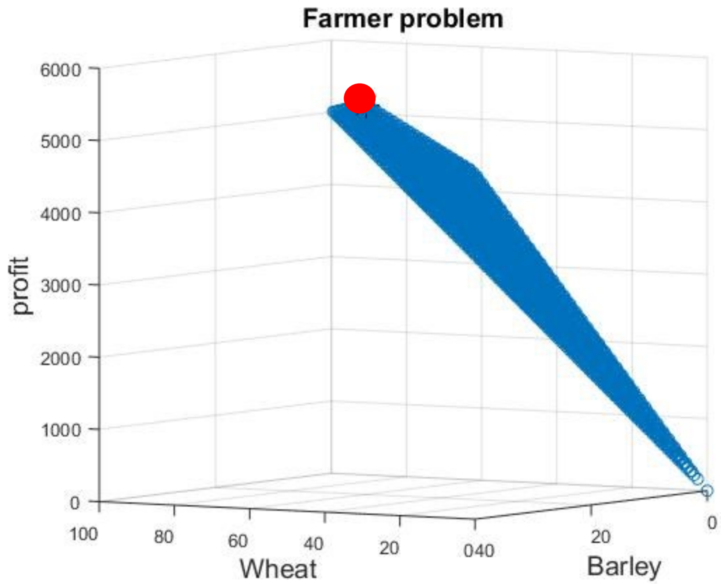
Graphical solution

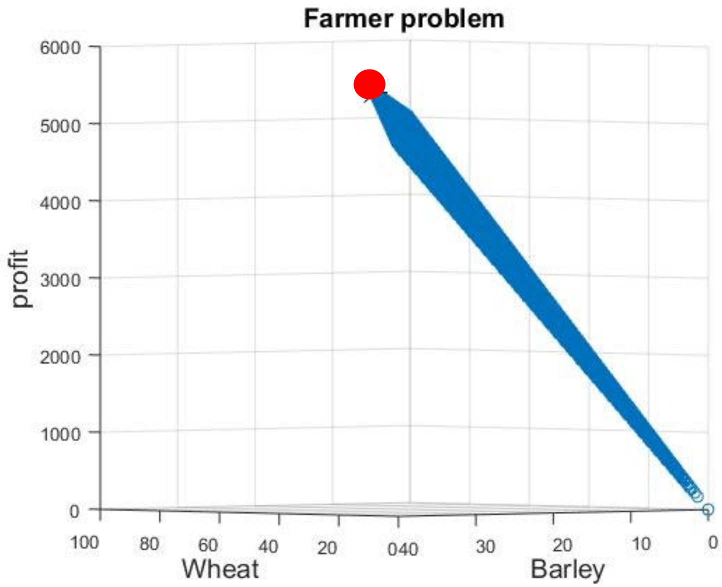


Solution – objective function

60 hectares of wheat
20 hectares of barley

Profit: $ 5400

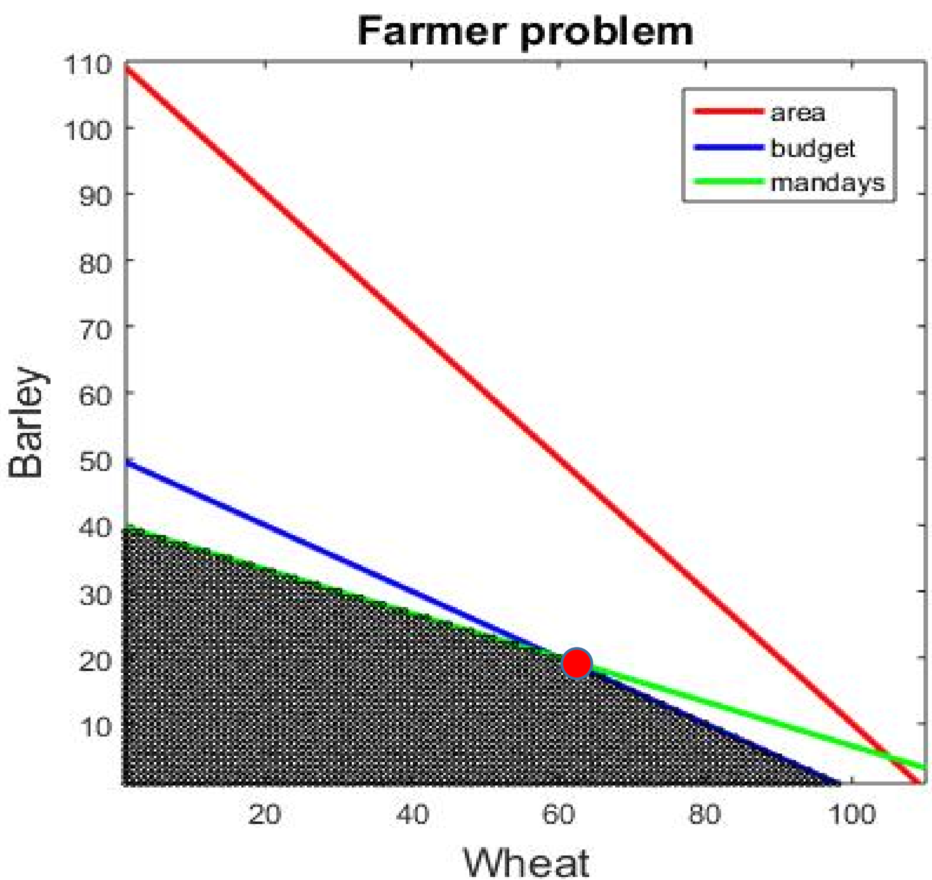




Same figure viewed from different angles

Solution

Note: solution will always be at the intersection between lines – possibly extending along a line



Massaging the problem

1. Some inequalities may be stated as

$$\mathbf{X_1 + 2\,X_2 \leq 100}$$

Others may be stated as

$$\mathbf{X_1 + 3\,X_2 \geq 120}$$

2. Some constraints may be equalities:

$$\mathbf{X_1 + X_2 = 10}$$

3. Optimisation may be maximisation or minimisation of the objective function. The solver we will use minimises.

4. Some x_i may not be ≥ 0

How do we handle these situations?

Massaging the problem

1. Some inequalities may be stated as

$$X_1 + 2\, X_2 \leq 100$$

Others may be stated as

Multiply with -1 to make $\geq$ into $\leq$

$$X_1 + 3\, X_2 \geq 120$$

2. Some constraints may be equalities:

$$X_1 + X_2 = 10$$

Replace with $\mathbf{X_1 + X_2 \leq 10}$ and $\mathbf{X_1 + X_2 \geq 10}$

3. Optimisation may be maximisation or minimisation of the objective function. The program we will use minimises.

Multiply objective function with -1 to make its maximum into a minimum

4. Some X_i may not be ≥ 0

Replace with $\mathbf{X_i = u - v;\ u \geq 0}$ **and** $\mathbf{v \geq 0}$

Example:

Massage the following linear programming problem

Maximize objective function:

$x_1 + 2x_2 + 3x_3 + 4x_4 + 5$

subject to the constraints:

$4x_1 + 3x_2 + 2x_3 + x_4 \leq 10$

$x_1 - x_3 + 2x_4 = 2$

$x_1 + x_2 + x_3 + x_4 \geq 1$

and

$x_1 \geq 0$, $x_3 \geq 0$, $x_4 \geq 0$ - Note no restriction on x_2

Solution:

Massage the following linear programming problem

Maximize objective function:

$x_1 + 2x_2 + 3x_3 + 4x_4 + 5$ + 5 is removed as it will be added to all solutions

subject to the constraints:

$4x_1 + 3x_2 + 2x_3 + x_4 \leq 10$

$x_1 - x_3 + 2x_4 = 2$ Replace by two inequalities

$x_1 + x_2 + x_3 + x_4 \geq 1$ Multiply by -1 to get $\leq$

and

$x_1 \geq 0, x_3 \geq 0, x_4 \geq 0$ - Note no restriction on x_2

Replace x_2 with u_2-v_2
and set $u_2 \geq 0$ and $v_2 \geq 0$ – in both
constraints and objective function

Solution:

Maximize $+ x_1 + 2u_2 - 2v_2 + 3x_3 + 4x_4$ subject to the constraints:

$4x_1 + 3u_2 - 3v_2 + 2x_3 + x_4 \leq 10$

$x_1 - x_3 + 2x_4 \leq 2$

$-x_1 + x_3 - 2x_4 \leq -2$

$-x_1 - u_2 + v_2 - x_3 - x_4 \leq -1,$

and

$x_1 \geq 0,\ u_2 \geq 0,\ v_2 \geq 0,\ x_3 \geq 0,\ x_4 \geq 0.$

How to solve the problem in MatLab?

Farmer's problem as matrices

$X_1 + 2\ X_2 \leq 100$: budget constraint

$X_1 + 3\ X_2 \leq 120$: man-day constraint

$X_1 + X_2 \leq 110$: area constraint

The same inequalies in matrix format:

A x ≤ b

$$\begin{pmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \leq \begin{pmatrix} 100 \\ 120 \\ 110 \end{pmatrix}$$

The objective function to be maximized: $f(X_1, X_2) = 50\ X_1 + 120\ X_2$:

$$f^T \cdot x = (50 \quad 120) \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$$

Linprog solves and also does some massaging for us

```
opts = optimoptions(@linprog,'Display','off','Algorithm','dual-simplex');
[x,fval,exitflag] = linprog(f,A,b,Aeq,beq,lb,ub,opts)
```

- A*x $\leq$ b are inequality constraints. A = [] and b = [] if no inequalities exist.
- Aeq*x = beq are the equality constraints.
- lb and ub are lower and upper bounds on x (remember to set lb=[0] otherwise your problem may be unbounded)
- Fval is the **minimum** value of the objective function
- Exitflag is 1 if a solution is found

Unbounded objective function

There may not be a maximum or minimum of our objective function with the given constraints

Linprog returns an exitflag = -3

No feasible solution

There may not be a solution that fulfils all our constraints, i.e. no feasible solution

Linprog returns an exitflag = -2

Exercises 1

Helping the farmer more

Farmer's wife intervenes

- Optimum solution uses 20 + 60 = 80 hectares. They have 110.
- They may therefore sell some hectares and increase the budget. The price of land is \$300 per hectare.

Exercise:

Modify the inequalities and find the new optimum conditions

Farmer's wife intervenes

Solution:

X_3 is the number of sold hectares. $X_3 \geq 0$ by definition.

New max profit is $5786 for 98.6 hectares of wheat, 7.1 hectares of barley and 4.3 hectares sold

New inequalities:

$X_1 + 2\,X_2 - 3X_3 \leq 100$ % budget constraint

$X_1 + 3\,X_2 \leq 120$ % man-day constraint

$X_1 + X_2 + X_3 \leq 110$ % area constraint

The objective function to be maximized is unaltered: $f(X_1, X_2) = 50\,X_1 + 120\,X_2$

MatLab needs a vector with 3 elements: [50 120 0]

X_3 does not influence the value of f directly – but represents some internal decision parameter. Such variables are termed *slack* variables.

Farmer’s father intervenes

Why not plant rye?

Variety	Cost (Price$/Hec)	Profit (Price$/Hec)	Man-days/Hec
Wheat	100	50	10
Barley	200	120	30
Rye	150	85	20

Exercise:

How does it look when rye is a possibility?

In this exercise land cannot be sold.

Not a unique solution when adding rye

Not detected by MatLab!

I got 40 hec of wheat and 40 of rye and the same profit of $ 5400 as without rye

- All the parameters for rye are a linear combination of those for wheat and barley (in this case the mean value)
- ½ hec of wheat + ½ hec of barley = 1 hec of rye

Variety	Cost (Price$/Hec)	Profit (Price$/Hec)	Man-days/Hec
Wheat	100	50	10
Barley	200	120	30
Rye	150	85	20

Not a unique solution when adding rye

Not detected by MatLab!

We can detect it from the determinant of A: det(A)=0

X_3 is the area with rye

$$\begin{pmatrix} 1 & 2 & \boxed{1.5} \\ 1 & 3 & 2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \le \begin{pmatrix} 100 \\ 120 \\ 110 \end{pmatrix}$$

Mean of
the two
first
columns

det() takes only a square matrix. That is why I have removed the selling of land from the problem

Duality

- Every linear programming problem has a dual problem.
- If the original problem is maximisation, the dual problem is minimisation.

Original problem:

$f^T x$ is to be maximised

$Ax \leq b$

All elements of $x \geq 0$

Transpose A
Interchange f and b

The dual problem:

$b^T y$ is to be minimised

$A^T y \geq f$

All elements of $y \geq 0$

$A^T y \geq f \Rightarrow -A^T y \leq -f$

Duality

- Every linear programming problem has a dual problem.
- If the original problem is maximisation, the dual problem is minimisation.

Original problem:

$f^T x$ is to be maximised

$Ax \leq b$

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Transpose A
Interchange f and b

The dual problem:

$b^T y$ is to be minimised

$-A^T y \leq -f$

All elements of $y \geq 0$

[x,fval] = linprog(-f,A,b,Aeq,beq,lb,ub,opts)

-f because linprog minimises

[y,fval] = linprog(b,-A',-f,Aeq,beq,lb,ub,opts)

Fval will take the same optimum numerical value

Duality of Farmer's problem

Original problem with selling of land:

$$f^T \cdot x = (50 \quad 120)\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \text{ to maximise}$$

$$\mathrm{A\,x} = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 3 & 0 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \leq \begin{pmatrix} 100 \\ 120 \\ 110 \end{pmatrix} = b$$

All elements of x ≥ 0

The dual problem:

$$b^T \cdot y = (100 \;\; 120 \;\; 110)\begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix} \text{ to minimize}$$

$$\text{-A}^T\,\mathrm{y} = \begin{pmatrix} -1 & -1 & -1 \\ -2 & -3 & -1 \\ 1 & 0 & -1 \end{pmatrix}\begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix} \leq -\begin{pmatrix} 50 \\ 120 \\ 0 \end{pmatrix}$$

All elements of y ≥ 0

Original problem:

$f^T x$ is to be maximised

$Ax \leq b$

All elements of $x \geq 0$

The dual problem:

$b^T y$ is to be minimised

$-A^T y \leq f$

All elements of $y \geq 0$

Original problem:

$$\mathbf{100\ X_1 + 200\ X_2 \leq 10000 \Rightarrow X_1 + 2\ X_2 \leq 100}$$

If we multiply a row in the original problem with constant, we change A and b – not f. We get the same x and therefore also same $f^T x$.

Dual problem:

Feeding the new A and the old f into the dual problem means that the value of y can be adjusted to make up for the change we did to A. Because we made the same change to b this making up will give the optimum value of b

We can multiply rows in the A and f of the dual problem as we wish and get the same solution of that problem – just like we could with the original problem.

Passing of a changed A and b from the original to the dual problem does not change the optimum value – but it scales the variable of the dual problem

Exercises 2